Levels of Data Abstraction

Summary Notes with Visual Structure

Levels of Abstraction

Definition

To simplify users' interaction with a database, complexity is hidden using **three levels of data abstraction**:

1. **Physical Level** – How data are actually stored, describing complex low-level structures.
2. **Logical Level** – What data are stored and the relationships among them, hides physical complexity (provides **physical data independence**).
3. **View Level** – Only part of the database is visible to users, simplifying interaction and providing security.

Note

Database administrators mainly work at the logical level, whereas end users interact primarily through views.

1. Physical Level

Definition

- Describes the actual storage of data and low-level structures.
- Users at this level are rarely concerned with details.

Note

Changes at the physical level do not affect logical or view-level users (**Physical Data Independence**).

2. Logical Level

Definition

- Represents the entire database in terms of simpler structures.
- Describes **entities, attributes, and relationships** abstractly.

Example

Example: Creating a table at the logical level

```
CREATE TABLE instructor (
    ID varchar(5),
    name varchar(20),
    dept_name varchar(20),
    salary number(8,2)
);
```

Note

Logical schema is used by application programmers and can usually be changed without affecting physical storage.

3. View Level

Definition

- Provides a tailored interface to the database for different users.
- Simplifies interaction and can enforce security constraints.

Example

Example: University clerk can view only student records but not instructor salaries.

Note

Multiple views can exist for the same database. Views hide logical complexity and sensitive information.

Schemas and Instances

Definition

Database Instance: The collection of information stored at a particular moment.
Database Schema: The overall design of the database, including structure and constraints.

Note

Analogy: Schema ~ variable declarations; Instance ~ current variable values.

Schemas at Different Levels

- **Physical Schema:** Overall physical structure of the database.
- **Logical Schema:** Overall logical structure, used by applications.

- **View Schema (Subschema):** Defines a specific view for user interaction.

Physical Data Independence

Definition

The ability to change the physical schema without affecting the logical schema or application programs.

Note

Applications depend on the logical schema. Interfaces between levels must be well-defined to minimize impact of changes.

Example

Traditional logical schemas were static. Modern applications may require flexible logical schemas where different records in the same relation may have different attributes.